

Formalization of Optimization Problem Templates

1 Linear Programming

1.1 Problem Form

Given $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, solve:

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & c^\top x \\ \text{subject to} & Ax \leq b \\ & x \geq 0 \end{array}$$

1.2 Difficulty Levels

The problem dimension (n, m) is determined by the difficulty level:

Level	n (variables)	m (constraints)
1	2	2
2	2	3
3	3	3
4	3	4
5	4	5
6	5	6
7	7	8
8	10	12

1.3 Generation

Coefficients are sampled from $\{-4, -3, \dots, 4\}$. A feasible point $x_0 \in \mathbb{Z}_{>0}^n$ is drawn first; constraints are set as $b = Ax_0 + s$ where $s \in \mathbb{Z}_{>0}^m$ is a random slack, guaranteeing feasibility. The objective weights c are derived to ensure a nonzero optimal value. The problem is solved with CLARABEL via CVXPY to produce the ground-truth label.

1.4 Template

Question:

Minimize the objective function $c^\top x$ where $c = [\dots]$, subject to the constraints $Ax \leq b$, where $A = [\dots]$ and $b = [\dots]$. What is the optimal value?

You must solve it using only mental mathematical reasoning. Do NOT write or execute any code. Do NOT use Python, MATLAB, Julia, or any programming language. Do NOT use CVXPY, scipy, numpy, or any solver library.

Answer: $c^\top x^*$ rounded to the nearest integer.

2 Quadratic Programming

2.1 Problem Form

Given $P \in \mathbb{R}^{n \times n}$ (positive semidefinite), $q \in \mathbb{R}^n$, $G \in \mathbb{R}^{m \times n}$, $h \in \mathbb{R}^m$, $A \in \mathbb{R}^{p \times n}$, $b \in \mathbb{R}^p$, solve:

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && \frac{1}{2} x^\top P x + q^\top x \\ & \text{subject to} && Gx \leq h \\ & && Ax = b \end{aligned}$$

2.2 Difficulty Levels

The problem dimension (n, m) is determined by the difficulty level, with $p = \lfloor n/2 \rfloor$ equality constraints:

Level	n (variables)	m (inequalities)	p (equalities)
1	2	2	1
2	2	3	1
3	3	3	1
4	3	4	1
5	4	5	2
6	5	6	2
7	7	8	3
8	10	12	5

2.3 Generation

Coefficients are sampled from $\{-4, -3, \dots, 4\}$. $P = M^\top M$ for a random integer matrix M , guaranteeing positive semidefiniteness. A feasible point $x_0 \in \mathbb{Z}_{>0}^n$ is drawn first; inequality constraints are set as $h = Gx_0 + s$ where $s \in \mathbb{Z}_{>0}^m$ is a random slack, and equality constraints as $b = Ax_0$, both guaranteeing feasibility. The problem is solved with CLARABEL via CVXPY to produce the ground-truth label. Up to 3 retries resample q , G , A if the solver does not return optimal.

2.4 Template

Question:

Minimize the quadratic objective $\frac{1}{2}x^\top Px + q^\top x$ where

$P = [\dots]$ and $q = [\dots]$,

subject to $Gx \leq h$ and $Ax = b$, where

$G = [\dots]$, $h = [\dots]$, $A = [\dots]$, $b = [\dots]$.

What is the optimal value?

You must solve it using only mental mathematical reasoning. Do NOT write or execute any code. Do NOT use Python, MATLAB, Julia, or any programming language. Do NOT use CVXPY, scipy, numpy, or any solver library.

Answer: $\frac{1}{2}(x^*)^\top Px^* + q^\top x^*$ (raw floating-point value).

3 Quadratically Constrained Quadratic Programming

3.1 Problem Form

Given $Q_0, Q_1, \dots, Q_m \in \mathbb{R}^{n \times n}$ (positive semidefinite), $c_0, c_1, \dots, c_m \in \mathbb{R}^n$, $d_1, \dots, d_m \in \mathbb{R}$, and radius $R > 0$, solve:

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && \frac{1}{2}x^\top Q_0 x + c_0^\top x \\ & \text{subject to} && x^\top Q_i x + c_i^\top x \leq d_i, \quad i = 1, \dots, m \\ & && \|x\|_2^2 \leq R^2 \end{aligned}$$

3.2 Difficulty Levels

Level	n (variables)	m (constraints)
1	2	1
2	2	2
3	3	2
4	3	3
5	4	3
6	5	4
7	7	5
8	10	7

3.3 Generation

$Q_i = M_i^\top M_i + \lambda I$ for random integer matrices M_i , guaranteeing positive definiteness (with $\lambda = 1$ for Q_0 , $\lambda = 0.5$ for constraints). A feasible integer point x^* is drawn; each d_i is set to $x^{*\top} Q_i x^* + c_i^\top x^* + s_i$ where $s_i \in \{1, 2, 3, 4\}$ is a random slack. The norm bound uses $R = \max(3, \|x^*\|_2 + 3)$. Coefficients are sampled from $\{-4, \dots, 4\}$. The problem is solved with CLARABEL via CVXPY; up to 3 retries resample all parameters.

3.4 Template

Question:

Consider the convex QCQP over $x \in \mathbb{R}^n$:

Minimize $\frac{1}{2}x^\top Q_0 x + c_0^\top x$
 subject to $x^\top Q_i x + c_i^\top x \leq d_i$ for $i = 1, \dots, m$,
 and $\|x\|_2^2 \leq R^2$.

$Q_0 = [\dots]$, $c_0 = [\dots]$, $Q_i = [\dots]$, $c_i = [\dots]$, $d_i = \dots$

What is the minimum value of the objective?

You must solve it using only mental mathematical reasoning. Do NOT write or execute any code. Do NOT use Python, MATLAB, Julia, or any programming language. Do NOT use CVXPY, scipy, numpy, or any solver library.

Answer: $\frac{1}{2}(x^*)^\top Q_0 x^* + c_0^\top x^*$ (raw floating-point value).

4 Semidefinite Programming

4.1 Problem Form

Given symmetric matrices $C, A_1, \dots, A_m \in \mathbb{R}^{k \times k}$, scalars $b_1, \dots, b_m \in \mathbb{R}$, and trace bound $\tau \in \mathbb{R}$, solve:

$$\begin{aligned} & \underset{X \in \mathbb{S}^k}{\text{minimize}} && \langle C, X \rangle = \text{tr}(C^\top X) \\ & \text{subject to} && \langle A_i, X \rangle = b_i, \quad i = 1, \dots, m \\ & && \text{tr}(X) = \tau \\ & && X \succeq 0 \end{aligned}$$

4.2 Difficulty Levels

Level	k (matrix size)	m (constraints)
1	2	1
2	2	2
3	3	2
4	3	3
5	4	3
6	5	4
7	7	5
8	10	7

4.3 Generation

$A_i = \frac{1}{2}(B_i + B_i^\top)$ for random integer matrices B_i . A PSD witness $X^* = MM^\top$ is constructed from a random integer matrix M , and right-hand sides are set as $b_i = \text{tr}(A_i^\top X^*)$, guaranteeing

feasibility. The trace constraint uses $\tau = \text{tr}(X^*)$. C is symmetrized from a random integer matrix; up to 3 retries resample C if the solver fails. All integer entries are drawn from $\{-4, \dots, 4\}$. The problem is solved with CLARABEL via CVXPY.

4.4 Template

Question:

Consider the semidefinite program over a symmetric $k \times k$ matrix X :

Minimize $\langle C, X \rangle$ subject to $\langle A_i, X \rangle = b_i$ for $i = 1, \dots, m$, $\text{tr}(X) = \tau$, and $X \succeq 0$.

$C = [\dots]$, $A_i = [\dots]$, $b_i = \dots$

What is the minimum value of $\langle C, X \rangle$?

You must solve it using only mental mathematical reasoning. Do NOT write or execute any code. Do NOT use Python, MATLAB, Julia, or any programming language. Do NOT use CVXPY, scipy, numpy, or any solver library.

Answer: $\text{tr}(C^\top X^*)$ (raw floating-point value).

5 Geometric Programming

5.1 Problem Form

A geometric program in standard form is:

$$\begin{aligned} & \textbf{minimize} && f_0(x) \\ & \textbf{subject to} && f_i(x) \leq 1, \quad i = 1, \dots, m \\ & && g_i(x) = 1, \quad i = 1, \dots, p \end{aligned}$$

where $x \in \mathbb{R}_{++}^n$, each f_i is a *posynomial* (a sum of monomials with positive coefficients), and each g_i is a *monomial*. A monomial has the form $c x_1^{a_1} \cdots x_n^{a_n}$ with $c > 0$ and $a_i \in \mathbb{R}$; a posynomial is a finite sum of monomials.

In our template, there are no equality constraints ($p = 0$), and the inequality constraints are:

$$f_0(x) = \sum_{k=1}^{t_{\text{obj}}} c_k \prod_{i=1}^n x_i^{\alpha_{ki}} \quad (\text{posynomial objective})$$

$$f_j(x) = \frac{1}{d_j} \sum_{k=1}^{t_{\text{con}}} c_k^{(j)} \prod_{i=1}^n x_i^{\beta_{ki}^{(j)}} \leq 1, \quad j = 1, \dots, m$$

$$g_{\text{coup}}(x) = \frac{1}{r x_p x_q} \leq 1 \quad (\text{coupling monomial})$$

$$g_i^+(x) = \frac{x_i}{5} \leq 1, \quad \frac{1}{2x_i} \leq 1, \quad i = 1, \dots, n \quad (\text{bounds } 0.5 \leq x_i \leq 5)$$

where $c_k, c_k^{(j)} > 0$ are positive integer coefficients, $\alpha_{ki} \in \{e_{\min}, \dots, e_{\max}\}$ with at least one negative exponent per objective monomial, and $\beta_{ki}^{(j)} \in \{0, \dots, e_{\max}\}$.

5.2 Difficulty Levels

Each level is parameterized by $(n, t_{\text{obj}}, m, t_{\text{con}}, e_{\min}, e_{\max})$ where n = variables, t_{obj} = objective terms, m = constraints, t_{con} = terms per constraint, and $[e_{\min}, e_{\max}]$ = exponent range:

Level	n	t_{obj}	m	t_{con}	$e_{\min}$	$e_{\max}$
1	2	2	1	1	0	2
2	2	2	2	1	0	2
3	3	2	2	2	-1	2
4	3	3	2	2	-1	2
5	3	3	3	2	-1	2
6	4	3	3	2	-1	3
7	4	4	4	3	-2	3
8	5	4	4	3	-2	3

5.3 Generation

A feasible reference point $x^* \in [1.2, 2.0]^n$ is sampled uniformly. Objective monomials have exponents in $[e_{\min}, e_{\max}]$ with at least one negative exponent (to prevent the GP from being trivially posynomial). Constraint bounds are set as $d_k = \left(\sum_j c_{kj} \prod_i x_i^{*\beta_{ij}} \right) \cdot \rho$ where $\rho \in (1.05 - 0.005\ell, 1.2 - 0.01\ell)$ is a tightness factor at level ℓ . A coupling constraint $1/(x_i x_j) \leq r$ links two randomly chosen variables. The problem is solved with SCS in GP mode via CVXPY.

5.4 Template

Question:

Find the minimum value of $f(x) = [\dots]$ subject to $[\dots]$, $0.5 \leq x_i \leq 5$, with $x_i > 0$. What is the minimum value?

You must solve it using only mental mathematical reasoning. Do NOT write or execute any code. Do NOT use Python, MATLAB, Julia, or any programming language. Do NOT use CVXPY, scipy, numpy, or any solver library.

Answer: $f(x^*)$ (raw floating-point value).